

STOR 155.010 Lecture 04

Numerical Data

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Centrality and Dispersion Measures

- **Centrality measures** provide insight into the “central” or typical values of numerical variables.
- Common centrality measures: **mean, median, mode**
- **Measures of dispersion** help gauge the spread of data (how stretched or squeezed) about a central value.
- Common measures of dispersion: **range, variance, standard deviation**

Mean

- The **mean** (or **average**) is calculated by summing all values of a numerical variable and dividing by the total number of observations.
- Highly sensitive to outliers.
- An appropriate measure of central tendency when the data is roughly **symmetrically** distributed.

Sample Mean

- The **sample mean** $\bar{x}$ is calculated as

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n},$$

where $x_1, x_2, \dots, x_n$ represent the n observed values.

- The **population mean** μ is also computed the same way. It is often not possible to calculate μ since population data are rarely available.
- The sample mean is a **sample statistic**, and serves as a **point estimate** of the population mean.
- Sample mean is a good estimate if the sample is good.

- The **median** is the value that separates the higher half from the lower half of the values in a variable.
- If the dataset has an **odd** number of observations, the median is the middle number. If it's **even**, the median is the average of the two middle numbers.
- Unlike the mean, the median is not affected by extreme values (outliers), i.e., it is a **robust** statistic.

Computing the median

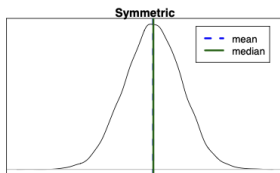
- 1 Order the numbers in increasing order.
- 2 If the multitude of numbers is odd, then the median is the number in the middle.
- 3 If the multitude of numbers is even, then the median is the average of the two middle numbers.

Example: Compute the mean and median of

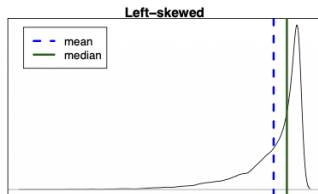
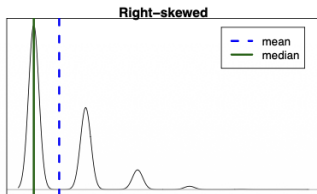
2, 5, 3, 8, 1, 6, 3, 2.

Mean vs. median

- If the distribution is **symmetric**, the center is $\text{mean} \approx \text{median}$.



- If the distribution is **skewed** or has extreme outliers, center is often defined as the median
 - Right-skewed: $\text{mean} > \text{median}$
 - Left-skewed: $\text{mean} < \text{median}$



Mean vs. Median: A Comparison

- **Sensitivity to Outliers:**

- Mean: Highly sensitive
- Median: Not sensitive (Robust)

- **Applicability:**

- Mean: Best for symmetric distributions
- Median: Useful for skewed distributions

- **Calculation Complexity:**

- Mean: Requires all data points
- Median: If the data is ordered, not all points are required

Mode

- The **mode** is the value that appears most frequently in a dataset.
- A dataset may have one mode (unimodal), more than one mode (multimodal), or no mode at all.
- The mode can be used for both numerical and categorical data.
- *Example:* The mode of $\{4, 2, 2, 3, 5, 4, 6, 2\}$ is 2, because it occurs three times, which is more than any other number in the dataset.

Q1 and Q3

- The 25th percentile is also called the first quartile, **Q1**. It is computed as the median of the ordered data to the left of the median.
- The 50th percentile is also called the median.
- The 75th percentile is also called the third quartile, **Q3**. It is computed as the median of the ordered data to the right of the median.

Example: Odd sample size

Data: 2, 4, 5, 7, 10

Example: Odd sample size

Data: 2, 4, 5, 7, 10

- Sample size = 5 (odd)
- Median (Q_2) = 5
- Lower half: 2, 4
- Upper half: 7, 10
- $Q_1 = \text{Median of Lower half} = \frac{2+4}{2} = 3$
- $Q_3 = \text{Median of Upper half} = \frac{7+10}{2} = 8.5$

Example: Even sample size

Data: 1, 2, 4, 5, 7, 10

Example: Even sample size

Data: 1, 2, 4, 5, 7, 10

- Sample size = 6 (even)
- Median (Q_2) = 4.5
- Lower half: 1, 2, 4
- Upper half: 5, 7, 10
- Q_1 = Median of Lower half = 2
- Q_3 = Median of Upper half = 7

Range

- The range is the difference between the largest and smallest values in the dataset. It provides a measure of total spread.
- Mathematically, the range of n numbers $x_1, \dots, x_n$ is defined to be the statistic

$$R = \max\{x_1, \dots, x_n\} - \min\{x_1, \dots, x_n\}.$$

- *Example:* The range of the 8 numbers $\{2, 5, 3, 8, 1, 6, 3, 2\}$ is
- Between $Q1$ and $Q3$ is the middle 50% of the data. The range these data span is called the **interquartile range**, or the **IQR**.

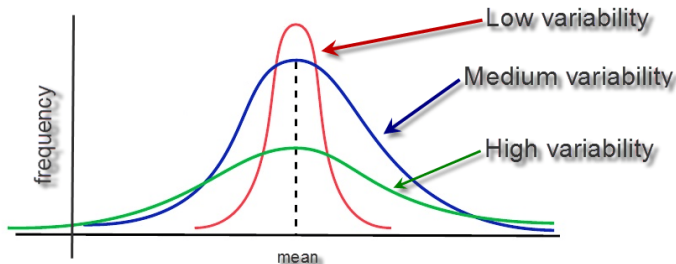
$$IQR = Q3 - Q1$$

Variance

Variance is roughly the average squared deviation from the mean.

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Example: Compute the variance of 2, 5, 3, 8, 7.



- Why use squares and not just take the average of the differences from the mean?
- In that case, we would always get

$$\begin{aligned}\frac{(x_1 - \bar{x}) + \cdots + (x_n - \bar{x})}{n} &= \frac{(x_1 + \cdots + x_n) - n\bar{x}}{n} \\ &= \frac{x_1 + \cdots + x_n}{n} - \bar{x} = 0,\end{aligned}$$

which is not informative.

Standard deviation

The **standard deviation** is the square root of the variance, and has the same units as the data:

$$s = \sqrt{s^2}$$

Example: Compute the standard deviation of 2, 5, 3, 8, 7.